

HAGI, Japan

WMO: 477540

Lat: 34.4100N

Lon: 131.4061E

Elev: 4

StdP: 101.28

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	0.0	0.8	-7.7	2.0	3.4	-6.5	2.2	4.1	10.3	4.3	9.1	4.7	4.0	170	0.390

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	6.8	33.2	25.5	32.0	25.6	31.0	25.4	26.7	31.0	26.2	30.5	25.7	29.9	4.1	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.6	20.9	29.3	25.0	20.1	28.8	24.5	19.5	28.4	84.0	31.1	81.7	30.6	79.6	30.1	28.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.4	7.1	6.1	-1.7	35.4	1.4	1.1	-2.7	36.2	-3.6	36.8	-4.4	37.4	-5.4	38.2		
			-3.2	27.5	1.5	0.6	-4.3	28.0	-5.2	28.3	-6.0	28.7	-7.0	29.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	16.1	5.9	6.5	9.2	13.8	18.4	22.0	26.5	27.7	23.5	18.2	13.1	8.2	(6)
(7)		DBStd	7.91	2.36	2.87	3.08	3.13	2.67	2.43	2.47	2.09	2.75	2.66	3.01	2.75	(7)
(8)		HDD10.0	371	131	106	54	6	0	0	0	0	0	6	69	(8)	
(9)		HDD18.3	1684	387	332	284	140	33	1	0	0	34	159	313	(9)	
(10)		CDD10.0	2605	2	7	27	119	260	361	511	547	405	253	99	14	(10)
(11)		CDD18.3	876	0	0	0	3	34	111	253	289	155	29	2	0	(11)
(12)		CDH23.3	7182	0	0	1	16	134	491	2368	3052	1020	99	2	0	(12)
(13)		CDH26.7	2329	0	0	0	2	15	90	808	1169	236	9	0	0	(13)
(14)	Wind	WSAvg	2.9	3.4	3.3	3.2	3.1	3.0	2.4	2.6	2.7	2.5	2.7	3.0	3.4	(14)
(15)	Precipitation	PrecAvg	1745	88	81	124	139	165	232	283	178	180	96	95	83	(15)
(16)		PrecMax	2418	184	124	241	288	349	466	649	548	379	221	222	166	(16)
(17)		PrecMin	1213	23	21	60	50	47	51	42	30	53	14	27	14	(17)
(18)		PrecStd	331	39	29	51	60	70	106	179	115	92	58	51	38	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.4	18.5	21.2	25.3	28.5	31.6	34.2	35.0	32.7	27.9	23.1	17.7	(19)
MCWB			12.0	14.8	13.5	16.2	19.0	24.1	25.7	25.8	25.0	22.0	17.4	13.6	(20)	
2%		DB	13.0	15.3	18.3	22.6	26.4	29.0	32.9	33.7	30.7	25.7	21.1	15.8	(21)	
		MCWB	9.5	11.6	13.1	16.0	18.1	22.7	25.6	25.5	24.7	21.0	16.4	12.2	(22)	
5%		DB	11.6	13.3	16.6	20.9	24.7	27.3	31.7	32.6	29.1	24.2	19.7	14.4	(23)	
		MCWB	8.2	9.9	12.3	15.2	17.9	22.4	25.5	25.7	24.2	19.6	15.7	10.8	(24)	
10%		DB	10.3	11.6	14.9	19.3	23.3	26.0	30.6	31.5	27.8	23.0	18.5	13.2	(25)	
		MCWB	6.8	8.3	11.2	14.4	17.7	21.9	25.4	25.7	23.4	18.4	14.7	9.9	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.1	15.0	15.8	19.7	22.0	25.2	27.2	27.6	26.3	23.8	19.1	15.3	(27)
MCDWB			14.6	17.7	18.1	22.6	25.0	29.6	31.2	31.5	30.4	26.9	21.0	16.5	(28)	
2%		WB	10.6	12.5	14.2	17.6	20.5	24.2	26.6	27.0	25.3	22.2	17.8	13.0	(29)	
		MCDWB	12.3	14.5	16.9	20.7	23.3	27.5	30.7	31.3	29.2	24.7	20.2	14.8	(30)	
5%		WB	8.7	10.5	13.0	16.2	19.4	23.5	26.1	26.5	24.7	20.7	16.6	11.6	(31)	
		MCDWB	10.9	12.5	15.7	19.4	22.7	26.3	30.3	30.8	28.1	23.2	19.0	13.8	(32)	
10%		WB	7.3	8.8	11.6	15.3	18.6	22.6	25.6	26.0	24.1	19.3	15.2	10.2	(33)	
		MCDWB	9.7	11.2	14.4	18.2	22.1	25.1	29.6	30.4	27.2	21.9	17.8	12.7	(34)	
(35)	Mean Daily Temperature Range	MDBR	6.3	7.1	8.3	8.7	8.3	6.4	6.2	6.8	6.9	8.1	7.8	6.8	(35)	
5% DB		MCDBR	8.8	9.9	10.9	11.2	10.6	8.2	7.7	8.1	7.8	9.0	9.0	8.8	(36)	
		MCWBR	6.1	7.0	7.1	6.3	5.1	3.8	3.0	2.9	3.6	4.7	5.5	6.0	(37)	
		MCDBR	7.8	8.2	8.9	8.4	7.8	6.3	7.1	7.4	6.7	7.2	7.4	7.5	(38)	
(39)		5% WB	MCWBR	6.3	6.8	6.8	5.5	4.7	3.4	3.1	3.3	3.4	4.3	5.2	6.0	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.439	0.483	0.551	0.577	0.576	0.592	0.591	0.549	0.509	0.464	0.441	0.431	(40)	
(41)		taud	2.017	1.941	1.792	1.784	1.842	1.889	1.938	2.061	2.112	2.124	2.102	2.051	(41)	
(42)		Ebn at Noon	728	742	726	733	740	726	723	748	757	755	724	708	(42)	
(43)		Edh at Noon	137	165	207	218	208	198	188	164	149	135	125	125	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.96	2.73	3.81	4.77	5.29	4.65	4.86	5.10	4.01	3.38	2.43	1.83	(44)	
(45)		RadStd	0.18	0.25	0.30	0.40	0.59	0.45	0.74	0.56	0.39	0.34	0.20	0.12	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(46)
(47)	Regional (25 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page